

Data Visualization & Insights Extraction

Univariate, Bivariate Analysis & Advanced Plotting

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Abstract

This presentation explores the foundational and advanced techniques of Data Visualization and Exploratory Data Analysis (EDA). Using the General Social Survey (GSS) dataset, we systematically break down data into actionable insights.

- **Univariate Analysis:** Examining continuous (Income, Age) and categorical (Gender, Education) distributions.
- **Bivariate Analysis:** Exploring relationships between variables (e.g., Income vs. Age, Height vs. Weight).
- **Advanced Plotting:** Utilizing Jointplots, Violinplots, and Faceting to uncover hidden patterns.

We focus on interpreting visual outputs, identifying outliers, and extracting real-world implications without relying on raw code syntax.

Session Objectives

- Master the art of summarizing continuous and categorical data using appropriate visualizations.
- Identify outliers, skewness, and data anomalies using boxplots and histograms.
- Evaluate bivariate relationships using scatter plots, regression lines, and correlation metrics.
- Understand the nuances of plotting libraries (Seaborn vs. Matplotlib) and troubleshoot common errors.
- Translate visual patterns into strategic, real-world business or research implications.

The General Social Survey (GSSsubset)

A comprehensive dataset capturing demographic, economic, and social indicators.

- **Sample Size:** $N = 3,000$ valid cases.
- **Variables (9):** id, sex, degree, income, marital, age, height, weight, hrswrk.
- **Data Types:** Continuous (Income, Age, Height, Weight), Categorical (Sex, Degree, Marital).
- **Research Goal:** To understand the distribution of wealth, health metrics, and social demographics within the surveyed population.

Part 1: Data Preparation & Structure

"Before we visualize, we must understand the shape of our data."

Data Importation & Dimensions

- Upon loading the `Clean_GSSsubset.csv` file, the first step is verifying the structural integrity of the dataframe.

Metric	Output
Dimension (<code>df.shape</code>)	(3000, 9)
Total Rows	3,000
Total Columns	9

- **Column Names:** `id`, `sex`, `degree`, `income`, `marital`, `age`, `height`, `weight`, `hrswrk`.
- **Implication:** The dataset is perfectly rectangular with no immediate structural anomalies. All 3,000 records contain the expected 9 features.

Variable Data Types

- Understanding data types dictates which statistical tests and visualizations are appropriate.

Data Type	Count	Variables
int64 (Integer)	3	id, age, hrswrk
float64 (Float)	3	income, height, weight
object (String/Cat)	3	sex, degree, marital

- Memory Usage:** 211.1+ KB (Highly efficient).
- Note:** height and weight are stored as floats, allowing for precise decimal measurements (e.g., 73.2 inches), which is crucial for accurate correlation analysis later.

Part 2: Univariate Analysis (Continuous)

"Understanding the shape, center, and spread of numerical data."

Income Distribution (Histogram)

- **Visualization:** Histogram with 50 bins, overlaid with a Kernel Density Estimate (KDE) curve.
- **Annotations:** A vertical red dashed line marks the **Median**.

Visual Insights

- Income typically exhibits a **right-skewed** distribution in socioeconomic data.
- The KDE curve helps visualize the underlying probability density, smoothing out the binning artifacts.
- Plotting the median is often more representative of the "typical" earner than the mean in skewed data.

Income Distribution (Boxplot & Outliers)

- **Visualization:** Vertical boxplot displaying the interquartile range (IQR), median, and mean.
- **Annotations:** Red line for Mean, Black dashed line for Median.

Outlier Detection

- By reloading a subset (`GSSsubset1.csv`), the boxplot explicitly flags data points falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.
- **Implication:** Extreme high-income earners pull the mean upward. The gap between the mean and median lines visually confirms the presence of positive skewness and high-value outliers.

Age Distribution (Histogram & Violin)

- **Side-by-Side Comparison:**

- ① **Histogram (bins=20):** Shows the raw frequency of ages. The red dashed line indicates the **Mean**.
- ② **Violin Plot:** Combines a boxplot with a rotated KDE. It reveals the *density* of the data at different age brackets.

Why use a Violin Plot?

While a boxplot only shows quartiles, the violin plot reveals if the age distribution is bimodal (e.g., a cluster of young adults and a cluster of retirees) or uniformly spread.

Joint Distribution (Height & Weight)

- **Visualization:** A Jointplot combining a central scatter plot with marginal histograms on the top and right axes.
- **Statistical Output:**

Metric	Value
Height and Weight Correlation	0.643

- **Interpretation:** A Pearson correlation of $r = 0.643$ indicates a **moderate-to-strong positive linear relationship**. As height increases, weight predictably increases.
- The marginal histograms confirm the individual distributions of height and weight within the joint context.

Part 3: Univariate Analysis (Categorical)

"Counting, comparing, and visualizing discrete groups."

Gender Distribution (Annotated Bar Plot)

- **Visualization:** Countplot (`sns.countplot`) with distinct color palettes and black edges for clarity.
- **Custom Annotation:** The exact percentage of each gender is calculated and annotated directly above the bars.

Research Implication

- Adding percentages transforms a simple frequency chart into an immediate demographic summary.
- If the dataset is meant to represent a national population, we can instantly check for sampling bias (e.g., is it exactly 50/50, or skewed?).

Educational Level Distribution (Vertical)

- **Visualization:** Vertical bar chart mapping the frequency of degrees (e.g., High School, Bachelor, Graduate).
- **Annotations:** Frequencies and percentages are annotated for immediate reading.
- **Observation:** The visual hierarchy immediately highlights the most common educational attainment in the sample.
- **Best Practice:** Categorical variables with long string labels (like "Junior College") are often better suited for horizontal bars to prevent x-axis label overlapping.

Educational Level (Horizontal Bar Chart)

- **Visualization:** Horizontal bar plot using `value_counts()`.
- **Design Choice:** The `viridis` color palette is used, with exact counts placed to the right of each bar.

Why Horizontal?

- **Readability:** Long labels like "LT HIGH SCHOOL" are fully readable without tilting the head or rotating text.
- **Ranking:** Horizontal bars naturally allow the eye to compare lengths from left to right, making it easy to spot the highest and lowest categories.

Marital Status (Pie & Count Plot)

- **Visualization:** A side-by-side figure featuring a Pie Chart and a Seaborn Countplot.
- **Technical Note:** The session encountered an `AttributeError` because Seaborn lacks a native `pie()` function.
- **Workaround:** The code successfully utilized Matplotlib's `axes[0].pie()` for the circular chart, while keeping `sns.countplot` for the categorical bars.

Implication

Knowing when to switch between Matplotlib (for geometric shapes like pies) and Seaborn (for statistical plots like bars) is a key data visualization skill.

Part 4: Bivariate Analysis

"Exploring relationships: How does one variable change with another?"

Income by Education Level (Violin Plot)

- **Visualization:** Violin plot with an inner boxplot, utilizing a coolwarm palette.
- **Axes:** X-axis = Education Levels, Y-axis = Income.

Visual Insights

- The "width" of the violin shows where most incomes are concentrated for each degree.
- **Expected Pattern:** Higher degrees (Graduate, Bachelor) will likely show violins shifted upward (higher median income) and potentially wider (greater variance in high-earning careers) compared to "LT High School".

Income by Gender (Bar Chart)

- **Visualization:** Categorical bar plot comparing aggregate Income across Gender categories.
- **Context:** This provides a high-level summary of the gender wage gap within the dataset.
- **Analytical Step:** While a simple bar chart shows the mean/median, it hides the distribution.
- **Best Practice:** Always pair a summary bar chart with a boxplot or violin plot to ensure the "average" isn't being heavily skewed by a few extreme outliers in one gender category.

Income by Education Qualification (Bar Chart)

- **Visualization:** Standard bar chart mapping Education Qualification to Income.
- **Utility:** Excellent for stakeholder presentations where simplicity is key.

Business Implication

- This chart directly answers the question: "What is the financial ROI of obtaining a higher degree?"
- If the bars show a steep incline from High School to Graduate degrees, it visually justifies educational investments or corporate tuition reimbursement programs.

Income vs. Age (Scatter & Regression)

- **Visualization:** Scatter plot with a superimposed linear regression line (`regplot`).
- **Aesthetics:** Alpha transparency (0.8) is used to handle overplotting, with a distinct red regression line.

Statistical Insights

- The regression line visualizes the trend: Does income generally increase with age (experience)?
- **Non-Linearity:** In reality, income often plateaus or drops near retirement age. A simple linear regression line might mask this. A lowess/non-linear regression curve might be more appropriate for age-income dynamics.

Part 5: Insights, Implications & Best Practices

"Moving from pretty pictures to actionable data science."

Key Insights: Income & Demographics

- **Skewness & Outliers:** The income histogram and boxplot confirm that wealth distribution is heavily right-skewed. Relying on the "Mean Income" for policy-making is misleading; the Median is required.
- **Education Premium:** The bivariate violin and bar plots visually quantify the "education premium," showing distinct income stratifications based on degree attainment.
- **Gender Dynamics:** Comparing income by gender requires careful outlier analysis, as extreme high-earners can distort the perceived wage gap in simple bar charts.

Key Insights: Health & Anthropometrics

- **Correlation Strength:** The jointplot yielded a Pearson correlation of **0.643** between height and weight.
- **Biological Plausibility:** This moderate-to-strong positive correlation aligns with biological expectations (BMI frameworks).
- **Data Quality Check:** If the correlation had been near 0, or negative, it would indicate severe data entry errors (e.g., height recorded in cm but mixed with inches, or weight swapped with another variable).
- **Action:** The jointplot's marginal distributions also allow us to instantly spot impossible biological values (e.g., a weight of 5 lbs or height of 90 inches).

Best Practices for Data Visualization

- ① **Always Annotate:** Adding means, medians, or percentages directly to the plot (as done with the income and gender charts) saves the audience from guessing.
- ② **Handle Overplotting:** Use alpha transparency (`alpha=0.8`) in scatter plots so dense clusters of points remain visible.
- ③ **Choose the Right Geometry:** Use Violin plots for distribution density, Boxplots for outlier detection, and Bar charts for simple categorical comparisons.
- ④ **Label Clearly:** Ensure axes have units (e.g., "Income (\$)", "Height (inches)").

Common Pitfalls to Avoid

- **The Pie Chart Trap:** Pie charts are notoriously difficult for the human eye to compare accurately. Use bar charts instead unless showing parts of a single whole (like the Marital Status example).
- **Ignoring the Margins:** When plotting bivariate scatter plots, always check the marginal distributions. Two variables might have a correlation, but if their individual distributions are heavily skewed, the correlation might be driven by three outliers.
- **Library Confusion:** Attempting to use Seaborn for non-statistical plots (like `sns.pie`) will result in `AttributeError`. Know the boundaries of your tools.

Advanced Techniques: Faceting

- **Concept:** Faceting (or small multiples) involves splitting a single plot into multiple subplots based on a categorical variable.
- **Example:** Instead of one massive scatter plot of Income vs. Age with overlapping colors for Gender, we create two side-by-side plots: one for Males, one for Females.

Why Facet?

- Eliminates color-blindness issues.
- Allows the viewer to see the *background distribution* and regression trends for each group independently without visual clutter.

The Role of EDA in Machine Learning

- **Feature Selection:** The correlation of 0.643 (Height/Weight) tells us we might not need both variables in a regression model; they carry redundant information (multicollinearity).
- **Data Transformation:** The right-skewed Income distribution indicates that before feeding "Income" into a linear regression or neural network, we should apply a Log Transformation to normalize it.
- **Handling Missingness:** The initial `df.info()` check ensures we know exactly how many null values exist before imputation strategies are applied.

Summary & Key Takeaways

- **Univariate Analysis** reveals the shape, center, and anomalies of individual variables (e.g., Income skewness).
- **Bivariate Analysis** uncovers relationships and correlations (e.g., Height vs. Weight, Income vs. Age).
- **Tool Mastery:** Seamlessly integrating Pandas, Matplotlib, and Seaborn allows for rapid, publication-quality insights.
- **Context is King:** A visualization is only as good as the domain knowledge applied to interpret it. Always ask: "What business or research decision does this chart support?"

Next Steps in Data Analysis

- ➊ **Multivariate Analysis:** Move beyond 2D plots to 3D scatter plots or pair plots (`scatter_matrix`) to view all correlations simultaneously.
- ➋ **Statistical Testing:** Use the visual differences observed (e.g., Income by Degree) to run formal ANOVA or Chi-Square tests.
- ➌ **Predictive Modeling:** Use the cleaned and visualized GSS dataset to train a model predicting Income based on Age, Education, and Hours Worked.

Questions?

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